

WEBSITE TRAFFIC ANALYSIS

Internship Project Report

Submitted as part of the Data Analytics Internship Program

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Submitted To

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Project Objective

To analyze website traffic data by cleaning and processing event logs, calculating key performance metrics, identifying user behavior patterns, and visualizing traffic trends to generate actionable insights and recommendations for improving website engagement and conversions.

Tools Used

- Python
- Google Colab
- Pandas
- Seaborn
- Matplotlib
- Data Visualization Techniques

Submission Date

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PROJECT REPORT

Introduction

The objective of this project was to analyze website traffic data and understand visitor behavior, user engagement, and website performance. By examining event-based website logs, the project aimed to identify traffic patterns, evaluate user interactions, and provide actionable recommendations to improve engagement and conversions. The analysis included data cleaning, feature engineering, KPI calculation, visualization, and insight generation.

Data Loading and Initial Exploration

The dataset was imported into Google Colab using Pandas and initially explored using functions such as `head()`, `shape()`, `info()`, and `describe()`. This helped in understanding the structure of the data, identifying missing values, checking data types, and reviewing the overall quality of the dataset.

The dataset contained information related to pageviews, clicks, previews, dates, countries, cities, artists, albums, tracks, and unique link identifiers.

Data Cleaning and Preprocessing

Several preprocessing steps were performed to improve data quality. Duplicate records were removed, missing values were handled, and categorical fields were standardized. The date column was converted into datetime format to support time-based analysis. Event categories were validated to ensure consistency, and text fields were cleaned to remove formatting issues.

These steps improved the reliability and accuracy of the dataset for further analysis.

Feature Engineering

Additional features were created from the date column, including Year, Month, Day, and Weekday. These features enabled deeper analysis of traffic patterns over time. Event distributions were also examined to understand the frequency of pageviews, previews, and clicks.

Key Performance Metrics

Several KPIs were calculated to evaluate website performance:

- **Estimated Sessions:** Calculated using unique link identifiers.

- **Estimated Users:** Approximated due to the absence of user IDs.
- **Bounce Rate:** Estimated using pages with pageviews but no further interactions.
- **Click-Through Rate (CTR):** Calculated by comparing clicks with pageviews.

Average session duration could not be calculated because timestamp-level session data was not available.

User Flow and Page Analysis

User interaction behavior was analyzed through an estimated flow:

Pageview → Preview → Click

This helped identify how visitors progressed through different stages of engagement. Entry page analysis identified the most visited landing pages, while estimated exit page analysis highlighted pages where users were likely leaving without further interaction.

Data Visualization

Data visualization was performed to better understand website traffic patterns, user engagement, and visitor behavior. Various charts and graphs were created using Python libraries such as Matplotlib and Seaborn to transform raw data into meaningful visual insights.

- User Flow Funnel
- Event Distribution Pie Chart
- Top Entry Pages Analysis
- Estimated Exit Pages Analysis
- Daily Traffic Trend
- Country-wise Traffic Analysis

These visualizations helped identify traffic trends, engagement levels, and optimization opportunities

Key Insights

Insight 1: High Traffic but Moderate Engagement

The website generated a large number of pageviews, but only a portion of visitors progressed to previews and clicks. This indicates that while traffic acquisition is strong, user engagement can be further improved.

Insight 2: Traffic Concentrated in Few Landing Pages

A small number of landing pages account for a significant share of total website traffic. Optimizing these pages can have a substantial impact on overall conversions.

Insight 3: Strong International Reach

Visitors originated from more than 200 countries, with Saudi Arabia, India, and the United States contributing the highest traffic volumes. These regions represent key markets for future growth.

Insight 4: Presence of High-Bounce Pages

Several pages received pageviews but generated little or no interaction. These pages may be causing user drop-offs and require optimization.

Insight 5: Click-Through Opportunities Exist

The click-through rate indicates that many users leave without taking action. Improved call-to-action design and content relevance could help increase conversions.

Recommendations to Improve Conversions for Alfido Tech

1. Optimize High-Traffic Landing Pages

Focus on the top-performing entry pages by improving page design, content structure, loading speed, and call-to-action placement. Even small improvements on these pages can significantly increase conversions.

2. Strengthen Call-to-Action (CTA) Elements

Use more visible and compelling CTA buttons such as "Get Started," "Request a Demo," or "Contact Us." Clear CTAs can encourage visitors to take the next step in their journey.

3. Reduce Bounce Rates on Underperforming Pages

Analyze pages with high pageviews but no clicks. Improve content quality, simplify navigation, and ensure that visitors can easily find relevant information.

4. Target High-Traffic Countries with Localized Content

Since Saudi Arabia, India, and the United States contribute a large share of traffic, creating region-specific content and marketing campaigns can improve engagement and conversion rates.

5. Conduct A/B Testing

Test different headlines, CTA placements, page layouts, and promotional messages. Continuous experimentation helps identify designs and content that generate the highest conversions.

Conclusion

This Website Traffic Analysis project examined user interactions, traffic distribution, landing page performance, and engagement patterns using website event logs. The analysis revealed that while the website attracts substantial traffic from multiple countries, user engagement and conversion opportunities remain underutilized. Key metrics such as estimated sessions, bounce rate, and click-through rate highlighted areas where visitors disengage during their journey. By optimizing high-traffic pages, improving call-to-action effectiveness, reducing bounce rates, and implementing targeted marketing strategies, Alfido Tech can enhance user engagement and improve overall conversion performance. The findings provide valuable insights that can support data-driven decision-making and future website optimization efforts.